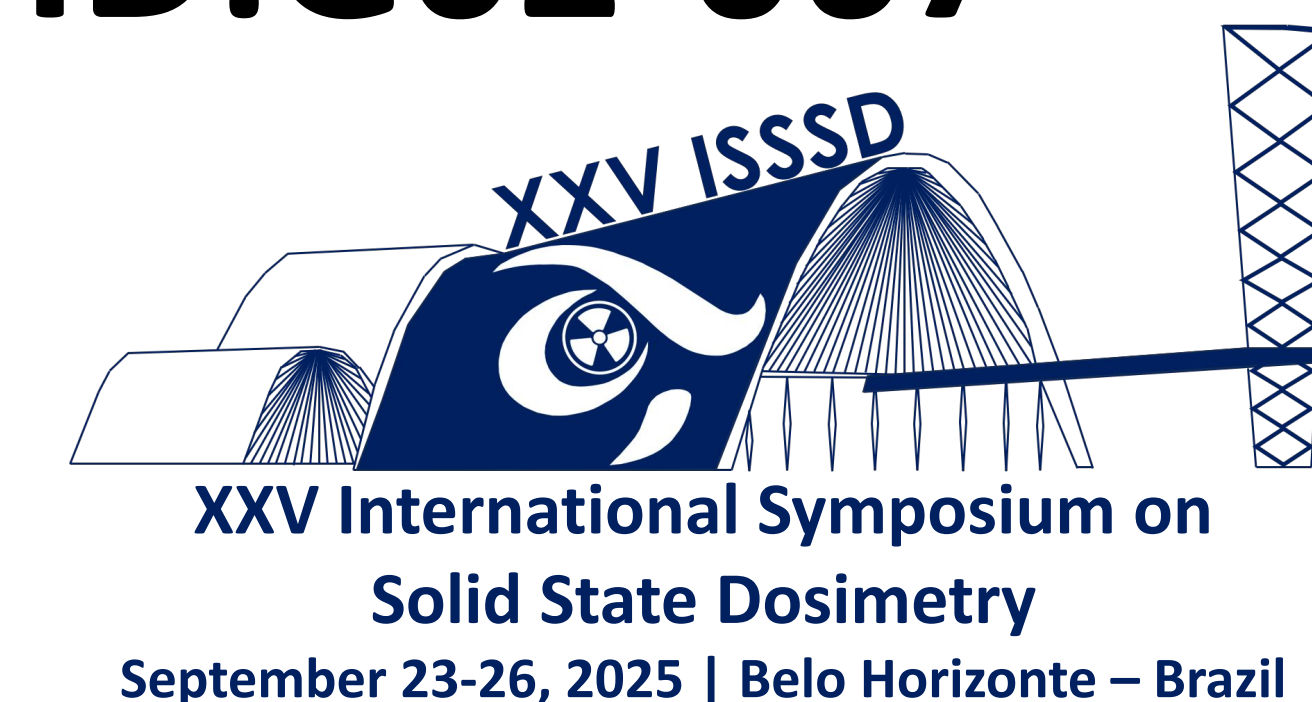


Monte Carlo Modeling of the BPW34 Photodiode for X-ray Dosimetry



A. C. H. de Oliveira¹, P. H. A. de Andrade^{2*}, J. W. Vieira^{2,3}

¹Pronto-Socorro Cardiológico Universitário de Pernambuco; ²Instituto Federal de Educação, Ciência e Tecnologia de Pernambuco; ³Universidade de Pernambuco

*Correspondence to: pedro.andrade@vitoria.ifpe.edu.br

@alexcholiveira



1. INTRODUCTION

The BPW34 is a general-purpose silicon PIN photodiode commonly used in optoelectronic applications due to its high spectral sensitivity, compact size, linearity, and affordability. Although originally developed for visible and near-infrared light detection, it has shown promising performance in detecting ionizing radiation such as X-rays and gamma rays.

Photodiodes like the BPW34 are known for high quantum efficiency, low dark current, and resistance to radiation-induced degradation, making them ideal candidates for radiation dosimetry. Experimental results from beam qualities N40 to N120 demonstrated high repeatability (<1.6%) and sensitivity, potentially due to its larger intrinsic layer and higher breakdown voltage [1].

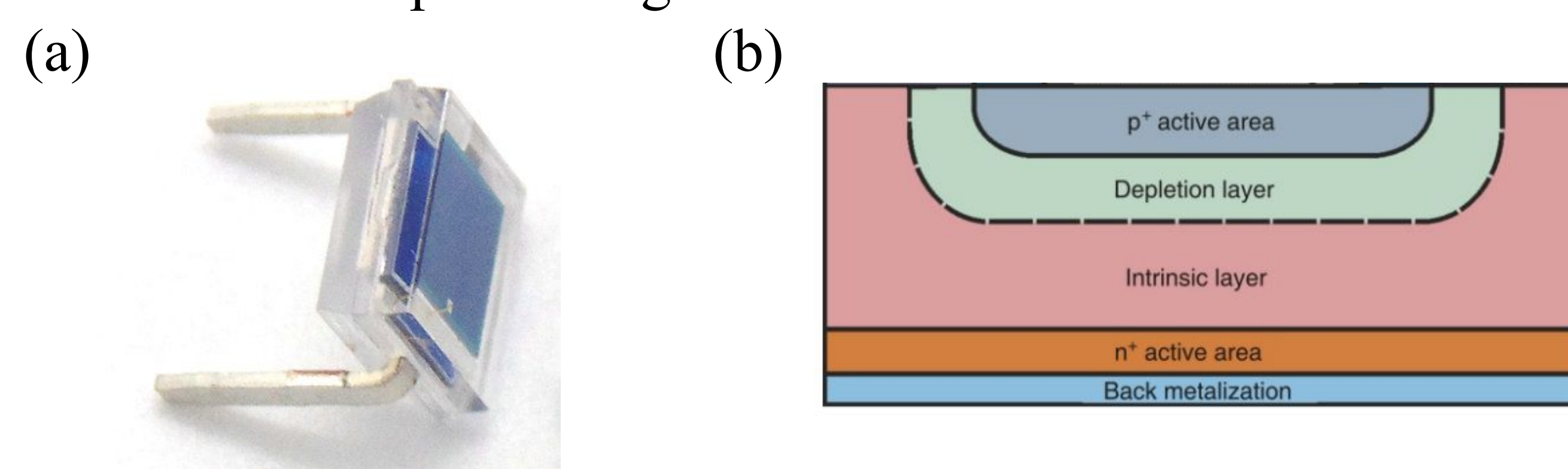
This study uses the Geant4 Monte Carlo toolkit to simulate the BPW34 and assess its dosimetric performance when exposed to X-ray beams in medical diagnostic ranges.

2. MATERIALS AND METHODS

A detailed 3D model of the BPW34 was implemented in Geant4 (v11.3.1), including: Epoxy encapsulation; Lead frame (3x3 mm², 300 μm thick [2]); P-type, N-type, and intrinsic silicon layers (total silicon thickness: 400 μm; depletion region: 10 μm [3]);

Figure 1(a,b) shows the BPW34 photodiode and the schematic representation of its structure, highlighting the active layers and depletion region implemented in the Geant4 model.

Figure 1. BPW34 photodiode (package) and schematic of its structure and depletion region.



Irradiation setup:

- Spectral X-ray beams (40–150 kV) generated via SpekPy v2 [4]
- Co-60 beam: 1.1732 MeV and 1.3325 MeV photons
- Beam fully irradiated the 4x4 mm² sensitive area

Validation was performed using ionization chambers A20 (X-rays) and A14 (Co-60) to simulate kerma in air, with the Standard EM Option 4 physics model and a cut-off range of 0.1 μm.

Charge collection modeling:

- Step-by-step energy deposition weighted by position-dependent diffusion factor $\eta(x)$

$$\eta(x) = \begin{cases} 1, & x \text{ inside the depletion region} \\ \exp\left(-\frac{d(x)}{L_{\text{side}}}\right), & x \text{ outside the depletion region} \end{cases}$$

- Diffusion lengths: $L_p = 500 \mu\text{m}$, $L_n = 1 \mu\text{m}$
- Absorbed dose = deposited energy / silicon mass
- Simulations performed with $1 \times 10^7/\text{mm}$ primary histories

3. RESULTS

- Absorbed dose uncertainties: 1% to 5%
- Application of 10 V reverse bias increased secondary ionizations by 43%
- Detection efficiency decreased with energy: from 4.7% (40 kV) to 1.3% (150 kV)

Table 1 – Comparison between simulated and experimental C/Gy values for different beam qualities.

Beam	Charge in Si (C)	Air Kerma (Gy)	C/Gy (Sim)	C/Gy (Exp) [1]	Error (%)
N40	6,91E-09	5,24E-04	1,321E-05	1,12E-05	-17,7
N80	2,01E-09	2,68E-04	7,508E-06	7,44E-06	-0,91
N120	1,18E-09	3,54E-04	3,327E-06	4,03E-06	17,4
Co-60	5,92E-09	3,88E-03	1,525E-06	1,72E-06	11,1

Differences from [1] may relate to the experimental setup (e.g., housing box). Further investigation would require larger irradiation area and simulation histories.

At N80, the application of reverse bias between 5 and 25 V increased the collected charge by 4.5% to 10.5%, reflecting the expansion of the depletion region [3] and the progressive saturation of charge collection efficiency.

Two empirical relationships were obtained: a quadratic fit relating charge to beam voltage (with/without copper filter) and a linear fit linking absorbed dose to generated charge. These allow dose estimation from measured charge, reinforcing the BPW34's potential as a calibrated, low-cost dosimeter.

4. CONCLUSIONS

The BPW34 photodiode shows promising capabilities as a low-cost, durable, and energy-resolving X-ray dosimeter. Monte Carlo simulations validate its detection efficiency, energy dependence, and response to applied bias. Empirical relationships derived from the simulations can support practical calibration procedures for clinical and radiological protection applications.

5. REFERENCES

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